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Quantum-Enhanced K-Nearest Neighbors for Text Classification: A Hybrid Approach with Unified Circuit and Reduced Quantum Gates

Amine Zeguendry, Zahi Jarir, and Mohamed Quafafou*

Text classification, a key process in natural language processing (NLP), relies on the k-nearest neighbors (KNN) algorithm for its simplicity and effectiveness. Traditional methods often grapple with the high-dimensional nature of textual data, leading to substantial computational demands. This study introduces a novel classical quantum k-nearest neighbors (CQKNN) algorithm, which integrates quantum circuits into a conventional machine-learning framework to enhance computational efficiency and reduce storage requirements. This hybrid approach uses a unified quantum circuit that simplifies multiple similarity calculations through mid-circuit measurements and qubit reset operations, significantly improving upon traditional multi-circuit quantum k-nearest neighbors (QKNN) models. The CQKNN algorithm, tested on datasets such as SMS Spam Collection, Twitter US Airline Sentiment, and IMDB Movie Reviews, not only outperforms classical KNN but also addresses challenges posed by noisy intermediate-scale quantum (NISQ) devices through advanced error mitigation techniques. This work highlights resource efficiency and reduced gate complexity and demonstrates the practical application of fidelity in quantum similarity calculations, setting new standards for quantum-enhanced machine learning and advancing current quantum technology capabilities in complex data classification tasks.

1. Introduction

In the last few years, the merging of quantum mechanics and information science has garnered increasing interest, leading to the rise of quantum information as an important field for study. Quantum machine learning (QML) arises as a highly prominent topic within the scientific topics. QML leverages the fundamental principles of quantum physics such as quantum entanglement and quantum superposition to perform computations.[1] QML, can be recognized as a quantum variant of conventional machine learning algorithms. It has the potential to reach an exponential or quadratic speed impact in quantum computing, while also demonstrating effective handling of quantum data. A variety of classical machine learning algorithms have been adapted to quantum versions, including quantum k-nearest neighbors (QKNN),[2–6] quantum neural network,[7–9] quantum support vector machine,[10,11] quantum regression algorithm,[12,13] quantum principal component analysis,[14] and so on.

Text classification is the process of assigning categories to textual data based on its content, a key task in natural language processing (NLP) with applications like sentiment analysis, spam detection, and intent detection. This process often employs machine learning techniques, both supervised and unsupervised, but faces challenges with large feature spaces. The k-nearest neighbors (KNN) algorithm[15] is widely used in text classification,[16] assigning classes to samples based on distance metrics. KNN has shown exceptional performance and has even been integrated with quantum computation, drawing significant research interest.

In 2014, Wieb et al. proposed a QKNN algorithm that uses the inner product distance[2] incorporating a squared acceleration effect. Additional important distance metrics include the Hamming, Mahalanobis, and polar distances. In 2017, Ruan et al. developed a QKNN algorithm employing the Hamming distance.[3] In 2022, Gao et al. introduced a modified version utilizing the Mahalanobis distance.[5] Subsequently, in 2023, Feng et al. presented a QKNN algorithm that incorporates the polar distance, covering both angle and module length data with the aid of an adaptable parameter.[6] Additionally, in 2024, Gong et al. proposed a divide-and-conquer strategy to further improve the efficiency of the QKNN algorithm for practical applications.[17] In the same year, Hdaib et al. discussed the integration of a QKNN algorithm within deep learning frameworks to enhance network security through effective anomaly detection.[18]

While the QKNN algorithms possess certain advantages, they also exhibit a number of limitations. The study discussed in ref.

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[2] focuses on the theoretical aspects of quantum phenomena that are not currently applicable in the context of quantum computing during the noisy intermediate-scale quantum (NISQ) era. In addition to the parameter k of QKNN algorithm, the method described in ref. [3] introduces a new parameter t. This parameter is utilized to identify the training samples for which the distance to the test sample is less than t. Consequently, the inclusion of this new parameter results in the need to set and optimize two parameters, thereby increasing the computational requirements. This is also the case for the algorithm discussed,[6] which incorporates the new parameter w. The algorithms described in refs. [4, 5] aim to load a dataset consisting of M data points into the circuit, requiring a larger quantity of qubits than is typically required. The vast majority of existing algorithms that aim to demonstrate the advantages of QKNN on real-world datasets share a common limitation, including the requirement for a large number of qubits and an increased number of circuit executions. To illustrate, when confronted with a dataset of M N-dimensions, certain algorithms require $\log_2(M+1)$ qubits, while others utilize N qubits. Additionally, a circuit is constructed M times for each test data point. Therefore, it is not feasible to handle a feature space with a high dimensionality using any of these algorithms.

However, existing QKNN implementations are often complex and resource-intensive, limiting their practical utility. Our proposed hybrid classical quantum k-nearest neighbors (CQKNN) algorithm introduces significant innovations to address these challenges. Our novel circuit design consolidates multiple similarity calculations into a single quantum circuit, drastically simplifying the quantum computing process and reducing the quantum resource requirements compared to traditional multi-circuit QKNN implementations. We also incorporate mid-circuit measurements and qubit reset operations, which enhance the efficiency of quantum resource utilization. These features are not yet commonly utilized in practical settings but enable the reuse of quantum states within the circuit. This capability minimizes the need for additional resources, facilitating more complex computations on NISQ devices.

Moreover, we move beyond theoretical discussions by practically implementing fidelity as a similarity metric. This demonstrates its effectiveness and scalability in real-world applications, providing a significant improvement over traditional methods. We also employ advanced error mitigation techniques to improve the reliability of our results, crucial for applications where noise and errors can significantly impact performance. The architecture of our CQKNN algorithm is poised to make a substantial impact in the field of quantum machine learning. By simplifying the quantum computation process and reducing resource demands, our approach not only makes quantum-enhanced algorithms more accessible but also sets a new standard for efficiency and practicality in real-world applications. This architecture addresses critical barriers that have hindered the broader adoption of quantum algorithms, offering a scalable and effective solution for text classification and potentially other complex data-driven tasks. Furthermore, the CQKNN requires only a logarithmic number of qubits, denoted as $\log_2(N)$ qubits to load an N-dimensional feature vector, and can be executed in a single time step for the entire process with $log_2(N) \times 2 + 1$ qubits in total.

Our innovations build upon recent advancements in quantum machine learning, yet they stand out by providing a practical framework that is applicable on today's quantum hardware. This makes our CQKNN algorithm a pivotal development in the ongoing integration of quantum technologies into mainstream computational tasks, promising significant advancements in both the theory and practice of quantum computing.

The remaining parts of this paper are structured in the following manner. In Section 2, we provide a comprehensive examination of the KNN classifier for text classification. Additionally, we present an overview of QML. Section 3 will serve as the main part of this paper, as it outlines our methodology and introduces our innovative circuit architecture designed for large-dimensional feature spaces. Section 4 presents the principal outcomes and evaluates the effectiveness of our CQKNN. For this evaluation, we will employ three distinct datasets: the SMS Spam Collection, Twitter US Airline Sentiment, and the IMDB Movie Reviews datasets for text classification. Moreover, we will compare the performance of our CQKNN against classical KNN. Section 5 of the paper discusses the conclusion and possible paths for future research.

2. Preliminaries

2.1. Text Classification

In this subsection, we provide an overview of the text classification (or text categorization) architecture and provide an explanation of the operational characteristics of each component involved in text classification. Text classification typically consists of three major components: data pre-processing, classifier development, and text classification, which will be discussed in detail below. Collectively, these three components make text classification practicable as illustrated in **Figure 1**. In this paper, we will concentrate on the red component to develop a classifier.

2.1.1. Data Pre-Processing

Data pre-processing is an essential phase in data analysis, consisting of six sub-components. These sub-components include document conversion, function word removal, word stemming, feature selection, dictionary construction, and feature weighting. The functioning of each component is described in the following manner:

1. Document converting: Transforms various document formats such as DOC, HTML, and PDF to typical text.
2. Function word removal: Eliminates topic-neutral words from the text, including articles (such as "a," "an," and "the"), prepositions (such as "in," "of," and "at"), conjunctions (such as "and," "or," and "nor"), and so on.
3. Word stemming: Is a process that aims to standardize the suffixes of words. For instance, it converts words like "labeling" to "label" and "introduction" to "introduct".
4. Feature selection: Is a technique used to decrease the dimensionality of the data space by eliminating features that are insignificant or less relevant.
5. Dictionary construction: Involves creating a standardized dictionary that serves as a point of reference for transforming a

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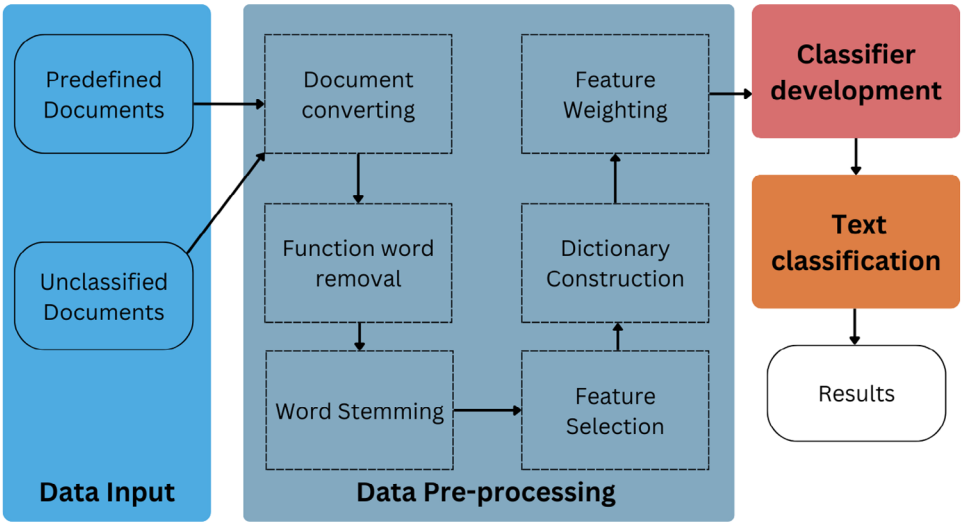


Figure 1. Architecture of text classification.

text document into a vector of features. Every element in the vector is related to a word from the dictionary.

6. Feature weighting: Uses the assignment of varying weights to words within a given dictionary. In our case, the weighting function employed is the standard normalized TF-IDF (Term Frequency-Inverse Document Frequency) algorithm, which uses word frequency to determine the significance of words within a document in relation to a larger collection of documents, known as a corpus.

Following these steps, each text will be transformed into a numerical feature vector, which can then be fed to the classifier.

2.1.2. Classifier Development

The development of a classifier is a crucial element in the process of automating text classification. The primary function of this component is to construct a classifier via the process of learning from existing documents. The resulting classifier is then utilized to classify documents of unknown class. The KNN algorithm depends on measurements of similarity. It has been extensively shown to be very efficient for tasks such as text classification and other problem fields. In the following subsection, a concise description of this algorithm will be presented.

2.1.3. Text Classification

The text classification component employs the model developed during the classifier development phase for the purpose of classifying new instances. In order for documents to be classified, they have to go through pre-processing, similar to the phase of developing the classifier.

2.2. KNN Algorithm

The KNN algorithm[15] is a commonly used supervised classification algorithm. The operational procedure of the algorithm is as follows: In the context of training and testing, it is frequently held that the training sample set comprises a total of M samples that are used for the purpose of training. Afterward, the distances between the test samples and the M training samples are computed. Subsequently, the k closest training samples are identified by comparing the results of this computation. If the majority of the training samples that are the k closest neighbors to a given test sample represent a certain class, then it may be assumed that the test sample also is related to the same class. The calculation of the distance between test samples and training samples is considered to be the most complex stage in the KNN algorithm. Furthermore, the computational complexity tends to have an increase in correlation with both the amount and dimension of the training data.

The classical process of KNN algorithm can be summarized as follows.

1. Initialize the value of k after loading the data.
2. Select a feasible distance metric and calculate the distance between the test sample and the set of M training samples.
3. Sort the calculated distances based on their distance values in ascending order.
4. Retrieve the top k elements from the sorted array.
5. Determine the class having the greatest occurrence among the k training samples, then assign that class as the classification for the sample in question.

Despite the fact that the KNN algorithm has superior performance and precision, it is crucial to select the appropriate distance metric. Typically, the Euclidean distance is used as the metric. The Euclidean distance, in practice, is a mathematical measure that combines the differences between two samples on each dimension, giving equal weight to all variables. However, this approach has some limits in relation to the usefulness of the data. In our work, we employ the cosine similarity, which is advantageous because even if two similar data objects are separated by a large Euclidean distance due to its size, they may still have a lower angle between them. Lowering the angle increases the similarity

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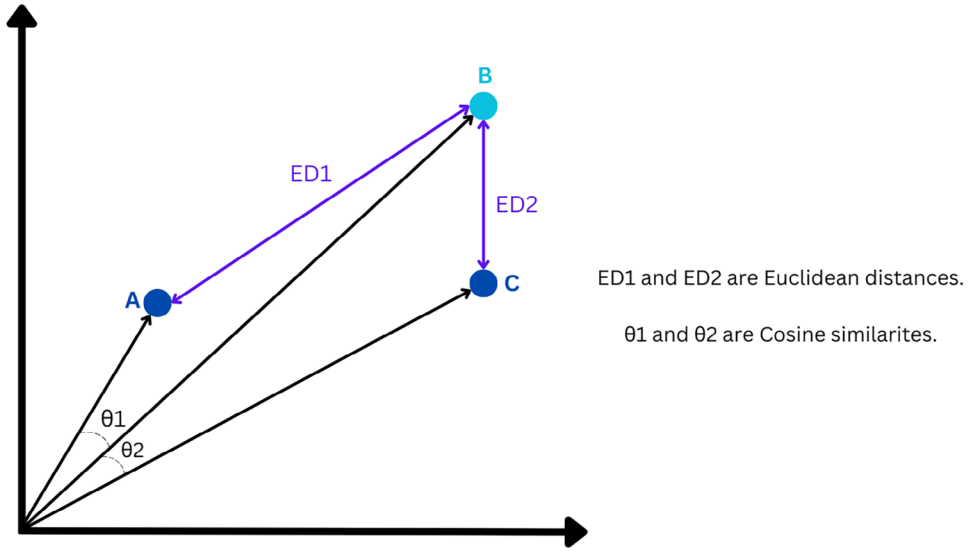


Figure 2. The difference between Euclidean distance versus Cosine similarity.

as shown in **Figure 2**. Cosine similarity is generally preferred over Euclidean distance for determining the similarity between texts of varying sizes.

2.3. Quantum Machine Learning

QML is the incorporation of quantum algorithms into machine learning applications. In this section, we will discuss briefly the various approaches to combining quantum computing and machine learning. We refer the reader to our previous work,[19] in which a comprehensive overview of quantum machine learning is presented.

In the field of integrating machine learning with quantum computing, four discrete approaches emerge. The effectiveness of these strategies relies on two primary factors: first, the source of the data, whether it was produced by classical (C) or quantum (Q) processes, and second, the type of computers used for data processing, whether classical (C) or quantum (Q) devices.[19] The categorization and its graphical representation are shown in **Figure 3**.

The CC scenario involves conventional data processing methods commonly used in conventional machine learning methods In this context, it refers to machine learning techniques that utilize strategies derived from quantum information research. Tensor networks, initially developed for quantum many-body systems, have been repurposed for training neural networks. Moreover, a range of machine learning algorithms with varying levels of quantum theoretical foundation exists, commonly referred to as 'quantum-inspired' algorithms.

The QC scenario investigates the potential integration of machine learning and quantum computing. Machine learning can play a crucial role in analyzing measurement data when we need comprehensive insights into the state of a quantum computing device, especially when the measurements are limited. This application is useful for differentiating quantum states produced by quantum sources and characterizing manipulations conducted by quantum experiments.

The CQ scenario utilizes quantum computing for the analysis of classical datasets. Data, including images and text, is transferred from classical computers to a quantum machine for analysis. The main aim of the CQ approach is to develop quantum algorithms specifically designed for data mining. The scientific community has proposed various solutions in this pursuit. The solutions could involve modifying existing machine learning algorithms to work well with quantum algorithms or developing completely new models that are inspired by the unique characteristics of quantum computing.

In the last QQ case, we examine the processing of "quantum data" by a quantum machine. Two possible interpretations emerge. The data could be generated via a quantum experiment using a system with quantum properties, and the resulting measurements would be inputted into a separate quantum computer. In a second interpretation, a quantum computer emulates the be-

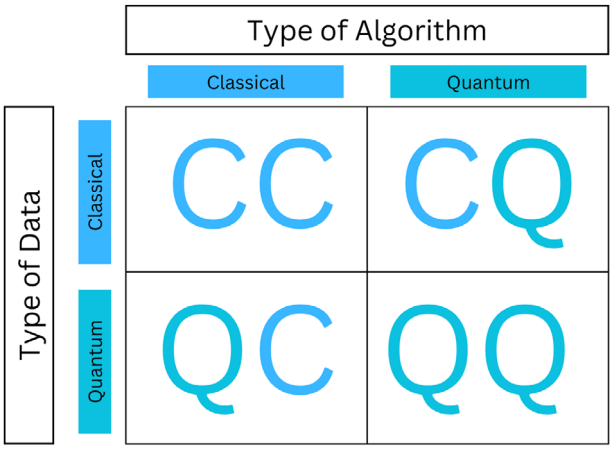


Figure 3. Four distinct approaches to integrating machine learning with quantum computing. Reproduced with permission from ref. [20] 2017, Maria Schuld.

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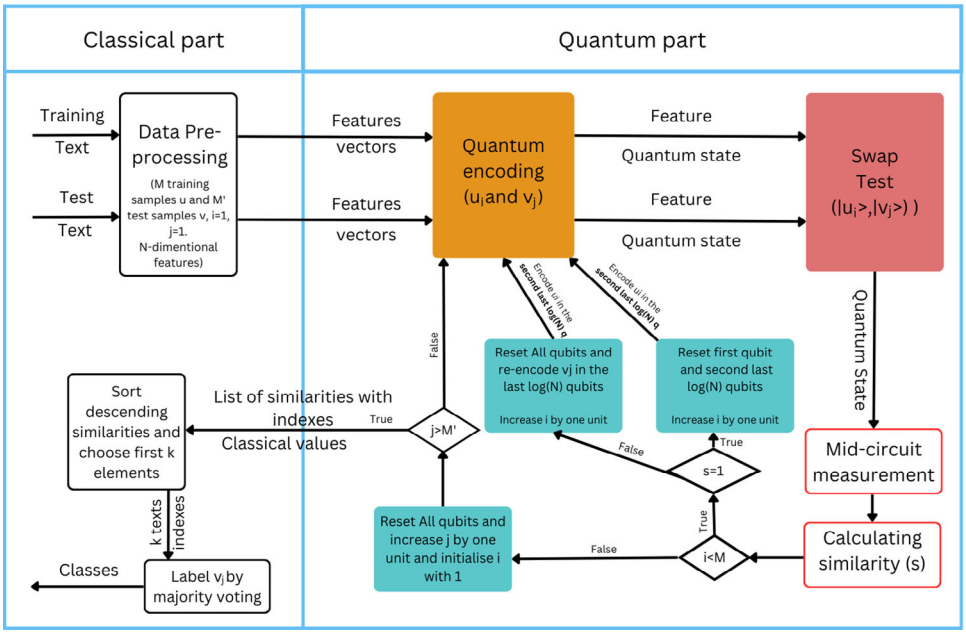


Figure 4. Diagram of our Hybrid CQKNN, with q representing qubits, and s the calculated similarity.

havior of a quantum system and utilizes the state of the system in question as input for a QML algorithm executed on the same quantum computer.

Our research focuses on the CQ scenario. Where we intend to develop a QML algorithm based on quantum subroutines, such as calculating similarity using fidelity between two quantum states and the method of measuring, resetting, and reusing qubits within a single quantum circuit. This approach involves the translation of conventional data into quantum data, which is commonly referred to as "quantum encoding", which we will cover in the next section.

3. The Proposed Methodology

Text classification is a systematic procedure that classifies a given text to a certain class based on predetermined criteria, ensuring a substantial level of certainty. The fundamental concepts of our text classification technique, which is based on CQKNN, are shown in **Figure 4**. Texts that have been classified serve as training samples $u = \{u_1, u_2, u_3, \ldots, u_M\}$, while the unclassified texts serve as the testing samples $v = \{v_1, v_2, v_3, \ldots, v_{M'}\}$. The classification of the test texts is based on training texts.

First, we used a traditional computer for pre-processing, feature extraction, and feature reduction in order to extract feature vectors from all the texts, as described in the preceding section. The feature vectors are then entered into a quantum device via the implementation of a method of quantum encoding for the generation of quantum states. The distance between the training state $|u_i\rangle$ $(1 \le i \le M)$ and the test state $|v_j\rangle$ $(1 \le j \le M')$, which is used to characterize their similarity, is then computed by the use of a quantum circuit, which operates in parallel and saved in classical bits using the mid-circuit measurement. Then, we reset the qubits in order to reuse them for calculating the similarity between subsequent data points and storing the distances in a list. The results will then be loaded into a traditional computer to obtain the k minimum similarities. Finally, indices of the k texts that are most comparable are determined, and majority voting is used to determine the final classification. The algorithm employed in the quantum part of our methodology is outlined as follows (Algorithm 1 and 2).

3.1. Quantum Part

In our work, we focused on the step of calculating distance or similarity in the KNN algorithm, which is the most important step of the algorithm. Therefore, we proposed a new quantum design to calculate the similarity between training and test samples. **Figure 5** illustrates the graphical frame design of the quantum circuit of our CQKNN algorithm. Our circuit contains four principal components: QE, Similarity, M, and QR. First, N represents the dimensionality of features, u is defined as the set of M training samples, and v represents the set of M' test samples, as previously discussed. QE, or quantum encoding, employs amplitude encoding to map high-dimensional text vectors onto quantum states using a reduced number of qubits. The 'Similarity' circuit is utilized to compute the similarity between two quantum states. In the red square, the 'M' denotes mid-circuit measurement, allowing real-time measurement of a single qubit during circuit execution. Qubit reset (QR) refers to the circuit utilized for qubit initialization. Detailed explanations of these components are provided in the following subsections.

3.1.1. Quantum Encoding

Numerous methodologies exist within the domain of Hilbert space to transpose classical data into quantum data. Fundamentally, the process of data encoding involves converting classical

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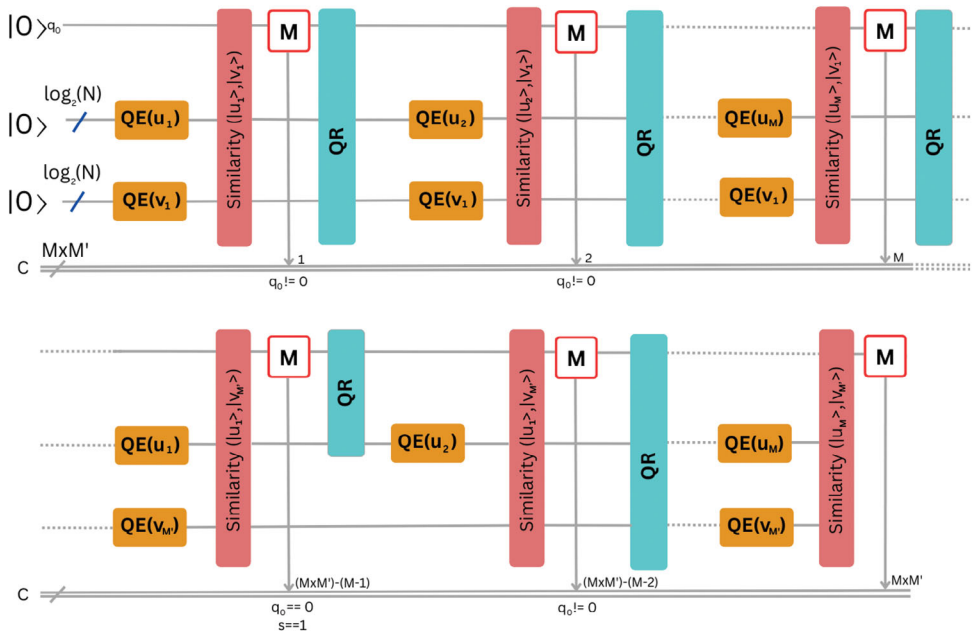


Figure 5. The proposed circuit architecture for our CQKNN's quantum part, where q_0 is the controlled qubit, N is the number of features, and s is the calculated similarity.

Algorithm 1 Quantum part of our CQKNN.

Require: u: M training samples (N-dimensional features)
Require: v: M' test samples (N-dimensional features)
Require: $log_2(N) \times 2 + 1$ qubits and $M \times M'$ classical bits
Require: q_0: Controlled qubit
for each v_j in V **do**
  for each u_i in U **do**
    Encode u_i in the second last $\log_2(N)$ qubits.
    Encode v_j in the last $\log_2(N)$ qubits.
    Apply the Swap test using q_0 as a controlled qubit.
    Measure q_0 and calculate the similarity s and store it in a list:
    if $s = 1$ **or** $q_0 = 0$ **then**
      Reset q_0 and the second last $\log_2(N)$ qubits.
      Encode the next training sample u_{i+1} in the second last $\log_2(N)$ qubits.
    else
      Reset all qubits.
      Encode the next training sample u_{i+1} in the second last $\log_2(N)$ qubits.
      Re-encode v_j in the last $\log_2(N)$ qubits.
    end if
    Repeat the Swap test and measurement steps.
  end for
  Proceed to the next test sample by incrementing the index j by 1.
end for
Retrieve similarities: After processing all test samples with the training samples, the list contains similarities and the indexes of the examples for each mid-circuit measurement.
returnList of similarities with indexes of samples.

Algorithm 2 Calculate Fidelity F Using Swap Test.

Require: $|\phi\rangle$: Quantum representation of an N-dimensional feature vector u on the initial three qubits.
Require: $|\psi\rangle$: Quantum representation of an N-dimensional feature vector v on the initial three qubits.
Require: $log_2(N) \times 2 + 1$ qubits
Require: q_0: Controlled qubit
Require: Number of measurements or shots M
1: **for** i **from** 1 **to** M **do**
2: Apply Swap Test to $|\phi\rangle$ and $|\psi\rangle$ with q_0 as the controlled qubit.
3: Get the measurement outcome (0 or 1) for the current iteration: M_i
4: **end for**
5: Count the total occurrences where M_i is 1: $T \leftarrow \sum_{i=1}^{M} M_i.count(1)$
6: Calculate the Fidelity:
7: $F \leftarrow 1 - \left(\frac{2}{M}\right) T$
8: **Output** F as the overall Fidelity between states $|\phi\rangle$ and $|\psi\rangle$

data into quantum states using a quantum computer.[19] Amplitude encoding emerges as the most effective method among these techniques for utilizing qubits in order to encode classical matrices or vectors. By establishing a fundamental connection between linear algebra and quantum computation, this method maximizes the utilization of quantum characteristics. The phenomenon is demonstrated by the amplitudes of the $|\psi_x\rangle$ quantum state, which is composed of n-qubits. It is mathematically represented as $|\psi_x\rangle = \sum_{i=1}^{N} x_i|i\rangle$, where $N = 2^n$ and represents a normalized classical d-dimensional dataset denoted by x. In this context, x_i represents the i-th element, and $|i\rangle$ represents the i-th computational base state. Significantly, x_i can comprise any type of number value, such as integer or floating-point values. To il-

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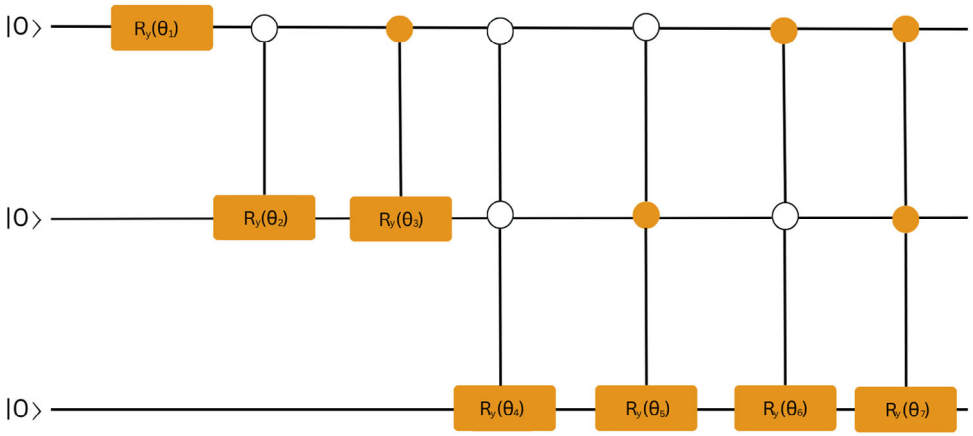


Figure 6. Three-qubit circuit for loading an 8D real vector into a quantum device using amplitude encoding.

lustrate, consider the representation of the four-dimensional array $x = (2.0, 1.0, -3.5, 0.0)$ using amplitude encoding. The first phase consists of normalizing the array, which yields the value $x_{norm} = \frac{1}{\sqrt{17.25}}(2.0, 1.0, -3.5, 0.0)$. In addition, the dimension of the vector should correspond to the value 2^n, where n represents the minimum number of qubits required to encode the vector. In case the vector lacks to satisfy this criteria, an additional zero is appended to it. Two qubits are adequate to encode the x_{norm} in this specific situation using the amplitude encoding principle as follows:

$$|\psi_{x_{norm}}\rangle = \frac{1}{\sqrt{17.25}}[2|00\rangle + |01\rangle - 3.5|10\rangle] \quad (1)$$

This state encodes the following matrix simultaneously:

$$X = \frac{1}{\sqrt{17.25}} \begin{pmatrix} 2.0 & 1.0 \\ -3.5 & 0.0 \end{pmatrix} \quad (2)$$

where the initial qubit denotes the row index, while the second qubit denotes the column index.

Then, the register of n qubits must be initialized with the value $|\psi_{x_{norm}}\rangle$. Starting from the highest position, the initial state is $|0 \ldots 0\rangle$, and the quantum circuit has been demonstrated **Figure 6**. The symbol R_y denotes a rotation operation performed on a single qubit around the Y-axis at an angle of θ. In the given context, the solid point indicates a control state of 1, while the empty point represents a control state of 0. The quantum circuit employs $O(N)$ sequentially applied multi-controlled gates with a circuit depth of $O(N)$. It is crucial to acknowledge that this depth exhibits variability in real-world quantum devices subsequent to the transpilation process, indicating a variable number of gates. This variability arises from the conversion of logical gates into multiple physical gates during the transpilation process. We will delve into this aspect in detail in the implementation section, providing examples to illustrate the effects of transpilation.

The majority of QKNN algorithms proposed in the literature employ other techniques to encode classical data to quantum states, such as basis encoding, which employs a large number of qubits to encode data. For instance, employing basis encoding requires N qubits when encoding N-dimensional real vectors. In contrast, only $\log_2(N)$ qubits are required by the method we have employed. Consequently, in our case study, a small number of qubits can be used to manipulate vectors with a high dimension.

3.1.2. Calculating the Quantum Similarity

Even with the advantage of quantum parallelization, retrieving information with high performance from a quantum state is a complex task. This is because, during the process of measuring a quantum state, the quantum state tends to partially collapse, and previous information obtained by quantum algorithms is lost. Buhrmann's technique (quantum fingerprinting)[21] is a straightforward solution to the problem of quantum state loss and enables the calculation of the similarity between two vectors with high precision and speed. His procedure is described as follows: The Hadamard gate takes the auxiliary qubit $|0\rangle$ and convert it to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. The circuit then uses a $SWAP$ gate to swap the vectors ψ and ϕ (i.e., $|\psi\rangle|\phi\rangle$ to $|\phi\rangle|\psi\rangle$). This superposition state performs the function of a control qubit to operate the swap gate in order to get $\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle|\phi\rangle + |1\rangle|\phi\rangle|\psi\rangle)$. This process is further illustrated in the circuit in **Figure 7**.

Quantum state is altered once the auxiliary qubit crosses through the H gate a second time:

$$|\beta\rangle = \frac{1}{2}|0\rangle(|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle) + \frac{1}{2}|1\rangle(|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle) \quad (3)$$

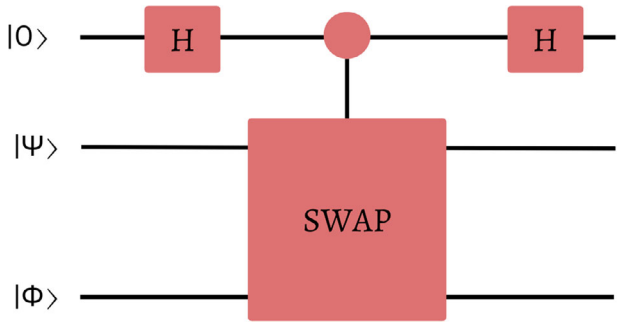


Figure 7. Swap test circuit to compute the similarity between ψ and ϕ.

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The probability that $\beta = 0$ is denoted by $P(0)$. Thus,

$$\begin{aligned} P(0) &= \langle\beta|\,|0\rangle\langle 0| \otimes I\,|\beta\rangle \\ &= \left[\frac{1}{2}\langle 0|\,(\langle\psi|\,\langle\phi| + \langle\phi|\,\langle\psi|) + \frac{1}{2}\langle 1|\,(\langle\psi|\,\langle\phi| - \langle\phi|\,\langle\psi|)\right]|0\rangle\langle 0| \\ &\quad \otimes I\left[\frac{1}{2}|0\rangle\,(|\psi\rangle\,|\phi\rangle + |\phi\rangle\,|\psi\rangle) + \frac{1}{2}|1\rangle\,(|\psi\rangle\,|\phi\rangle - |\phi\rangle\,|\psi\rangle)\right] \\ &= \left[\frac{1}{2}\langle 0|\,(\langle\psi|\,\langle\phi| + \langle\phi|\,\langle\psi|)\right]\left[\frac{1}{2}\langle 0|\,(|\psi\rangle\,|\phi\rangle + |\phi\rangle\,|\psi\rangle)\right] \\ &= \frac{1}{4}(\langle\psi|\,\langle\phi|\,|\psi\rangle\,|\phi\rangle + \langle\phi|\,\langle\psi|\,|\phi\rangle\,|\psi\rangle + \langle\psi|\,\langle\phi|\,|\phi\rangle\,|\psi\rangle \\ &\quad + \langle\phi|\,\langle\psi|\,|\psi\rangle\,|\phi\rangle) \end{aligned} \tag{4}$$

The feature vectors are normalized such that $\langle\phi|\phi\rangle = 1$ and $\langle\psi|\psi\rangle = 1$. As a result,

$$\begin{aligned} P(0) &= \frac{1}{4}(1 + 1 + \langle\psi|\,\langle\phi|\,|\phi\rangle\,|\psi\rangle + \langle\phi|\,\langle\psi|\,|\psi\rangle\,|\phi\rangle) \\ &= \frac{1}{4}(2 + 2|\langle\phi|\psi\rangle|^2) \\ &= \frac{1}{2} + \frac{1}{2}|\langle\phi|\psi\rangle|^2 \end{aligned} \tag{5}$$

Where the absolute value of the inner product $|\langle\phi|\psi\rangle|^2$ indicates the "fidelity" between the states $|\phi\rangle$ and $|\psi\rangle$. This measure is utilized to quantify the similarity between the two states, similar to the use of cosine similarity in conventional data analysis. We utilize $P(0)$ to symbolize the similarity between vectors, represented as $s(u_i, v_j)$. If both quantum states were orthogonal, the fidelity is 0; if they are identical, the fidelity is at its highest, which is 1. Accordingly, an increase in the fidelity of two states corresponds to a proportional increase in their similarity s. The dimension of the quantum state vectors determines the efficiency of the quantum solution; as the dimension of the state vectors increases, the quantum solution that employs fidelity becomes more efficient. This is the rationale behind our selection of this metric, to compute the similarity in our CQKNN in order to use it for large-dimensional features, such as text classification. The algorithm for computing the fidelity F is outlined below, thus expressing the similarity $s(u_i, v_j)$ as:

$$s = \frac{1}{2} + \frac{1}{2}F \tag{6}$$

The CQKNN algorithm innovatively enhances the traditional KNN approach by implementing fidelity calculations through quantum computing, notably using the Swap Test across multiple iterations or shots. This dynamic measurement strategy not only bolsters the statistical accuracy of the fidelity estimation but also seamlessly integrates quantum statistical mechanics into machine learning frameworks, allowing the algorithm to surpass the performance of classical KNN methods. These quantum enhancements enable the CQKNN to operate with higher precision and speed, especially effective when handling complex datasets that pose significant challenges to conventional algorithms. This novel integration highlights the potential of quantum computing to transform and advance the field of machine learning by leveraging unique quantum properties. This approach, while theoretically proposed and robustly designed to fit within existing quantum computing frameworks, has not yet been implemented in the literature, marking our method as a pioneering contribution that could set a precedent for future quantum-enhanced machine learning applications.

3.1.3. Mid-Circuit Measurement and Qubit Reset

The amount of available qubits is a constraint for many quantum computers currently in use. In order for quantum computers to outperform their classical counterparts in a variety of practical applications, qubits will need to be utilized effectively. Consequently, optimization of circuits is an essential factor in quantum information science and applications. The authors of ref. [22] therefore proposed the "qubit" reuse technique, which presents an automated technique for the compilation of circuits, aiming to optimize their execution on a reduced number of qubits. This is achieved by mapping a given circuit onto a compressed variant of itself via the use of mid-circuit measurement and reset techniques. Mid-circuit measurement is a device feature accessible through some providers of quantum hardware. This effective feature enables the measurement of a single qubit in real-time during circuit execution, as opposed to measuring all qubits only after execution is complete.

Qubit reset is a crucial mechanism in quantum computing that addresses the resource constraints of NISQ devices, where the availability of qubits is limited. This process involves returning a qubit to a known standard state, typically the ground state, after it has been used in a calculation. This capability is vital for enabling repeated use of qubits within the same quantum circuit, thus maximizing computational throughput without requiring a proportional increase in physical qubits. By reinitializing qubits to their baseline state, quantum systems can undertake additional operations without the need to start from the beginning, significantly improving the efficiency and scalability of quantum algorithms.

Combining mid-circuit measurement with qubit reset presents a promising avenue for recycling quantum resources, thereby enhancing device efficiency by enabling the reuse of qubits. This leads to the optimization of quantum circuits, as illustrated in **Figure 8**, in which the qubit state is reset after interactions on that qubit have concluded and a measurement has been conducted. The reset qubit can then store the state and perform the interactions of another qubit in the quantum circuit.

In our study, we implemented the qubit reuse technique in order to reuse qubits following the measurement of the calculated similarity between each training sample and test sample. As shown in Figure 5, the outcomes of each measurement are recorded in conventional registers c_i, after each similarity measurement we reset the qubits used to store the quantum states that represent our data. Furthermore, the results will be transmitted to the traditional computer, which controls the traditional portion of our algorithm. By integrating qubit reuse, the CQKNN architecture not only tackles some of the biggest challenges in quantum computing but also sets a new standard for practical quantum-enhanced machine learning applications. Remarkably, these techniques have not been previously implemented in

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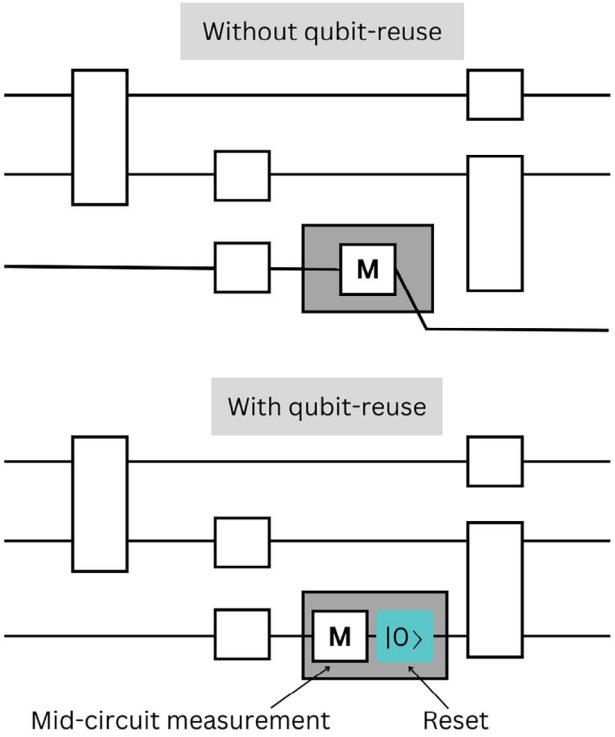


Figure 8. On the first circuit, the third input qubit is not used until after crossing the grey region, while this qubit is no longer used after that region. The overall circuit can be executed using one fewer qubit using mid-circuit measurement and reset, as depicted on the second circuit.

real-world scenarios, making our approach a ground-breaking effort in applying quantum mechanics to practical machine-learning problems.

To enhance the understanding of the circuit shown in Figure 5, an illustrative example is given. A dataset consisting of M training samples (u) and M test samples (v). Each sample in the dataset is characterized by 8D features. To effectively handle an 8D feature vector within our circuit, a logarithmic base-2 number of qubits, namely $\log_2(8)$, is sufficient. As a result, a total of three qubits have been assigned for training samples, and an additional three qubits are assigned for test samples in order to encode them. Furthermore, the first qubit experiences the application of a Hadamard gate, transitioning it into a superposition state. Subsequently, the measurement of this qubit occurs after the calculation of the similarity between the training samples and test samples. In addition, it is necessary to use a classical register of size $M \times M'$ to store the computed similarities. Upon performing the computation to determine the quantum similarity $s(u_1, v_1)$ between the initial training sample (u_1) and the initial test sample (v_1), the result will be stored in the primary classical register. Afterward, the algorithm verifies if the value of $s(u_1, v_1)$ is equal to 1 or not. In the case of $s(u_1, v_1) = 1$, the algorithm resets the first qubit and the second last three qubits, using them to code the next training sample u_2. The algorithm calculates the similarity between u_2 and v_1, where v_1 is already encoded in the last three qubits. In the second scenario, the algorithm performs a reset operation on all qubits, in order to encode the next training sample and re-encode v_1. After calculating the similarity between the quantum states u_2 and v_1, the algorithm continues this process until it calculates the similarity between u_M and v_1. It then resets all the qubits and repeats the same procedure to calculate the similarity between all M training samples and M' test samples.

The process of resetting qubits for subsequent use is an essential component of our quantum algorithm. The use of this strategic technique enables the optimization of efficiency in quantum computing while maintaining accuracy. In a systematic way, after each calculation, the qubits are reset to ensure their availability for the essential subsequent encoding and computations. The optimization of quantum computing processes not only enhances their efficiency but also facilitates the accurate and efficient handling of complex calculations in the field of quantum machine learning.

This entire procedure utilizes a single circuit with the fewest possible qubits, which will be executed only once. Currently, all proposed QKNN algorithm employs a circuit that is executed M times for each test data point and $M \times M'$ times overall to compute the similarity between training and test samples. Our approach significantly reduces the number of gates involved.

4. Experiments and Results

In our study, we evaluated several distance metrics for the K-Nearest Neighbors algorithm, including Mahalanobis, Manhattan, Euclidean, and cosine similarity. While the Mahalanobis and Manhattan metrics have their advantages in specific scenarios, Euclidean and cosine similarity demonstrated the highest reliability and performance across our tests. Due to their effectiveness in managing the complexities of text classification, we focused our detailed analysis on these two metrics and compared them with our CQKNN method to underscore the enhancements provided by quantum computing.

Additionally, while we initially considered comparing our CQKNN with other quantum KNN models, we found these models to be mostly theoretical and challenging to implement, particularly in text classification tasks. Such models typically require substantial increases in qubits, complex circuit setups, and multiple executions, rendering them impractical on current NISQ devices and generally limited to simulations. We also explored implementing our CQKNN without qubit reuse techniques; however, this approach proved too resource-intensive for effective execution on existing quantum computers. Consequently, our comparative analysis primarily focused on the efficiency and resource requirements of our method, emphasizing its reduced dependency on quantum resources and its practical feasibility within the constraints of today's quantum technology.

4.1. Input Dataset

The limitations intrinsic to current NISQ devices impose significant constraints on the deployment of QML techniques. To address this, our study utilizes well-established datasets characterized by a restricted number of features to emulate real-world text classification scenarios:

1. SMS Spam Collection Dataset: This dataset includes SMS labeled as spam or ham, providing a binary classification framework ideal for testing spam detection algorithms.[23]

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2. Twitter US Airline Sentiment Dataset: Contains tweets related to US airlines, originally classified into three categories: positive, neutral, and negative.[24] For our analysis, we specifically focus on the binary classification of tweets as either positive or negative to streamline the sentiment analysis process.
3. IMDB Movie Reviews Dataset: Comprises movie reviews with binary sentiment labels, making it a robust dataset for testing natural language processing and sentiment classification techniques.[25]

These datasets allow us to assess the performance of our CQKNN algorithm across different feature dimensions and highlight its application potential in diverse areas such as spam detection, sentiment analysis, and natural language understanding.

4.2. Data Pre-Processing

After importing the datasets, the initial step involves pre-processing the text data, a crucial stage for any NLP application. This process transforms raw text into a structured format ready for further analysis, as outlined in Section 2. Text pre-processing includes several steps: lexical analysis (which involves word tokenization, removing punctuation and special characters, and case normalization), eliminating stop-words, and applying lemmatization. These steps are essential for creating accurate document representations that are critical for the effectiveness of text classification applications.

For feature extraction, we utilize the TfidfVectorizer from the scikit-learn library, which converts text data into a TF-IDF matrix of features. This matrix, limited to a maximum of 32 features, forms the input for both our CQKNN and the classical KNN models. To accommodate the computational limitations of current quantum hardware, we reduced our datasets to 50 samples. The samples, along with their labels, are then divided into training and testing sets with an 80–20% split, respectively. This setup is applied across the three datasets utilized in our study: the SMS Spam Collection, Twitter US Airline Sentiment, and IMDB Movie Reviews datasets, each providing a unique challenge and context for evaluating the performance of our proposed methods.

4.3. Implementation

In our experimental setup, we evaluated four distance metrics using the Scikit-Learn "KNeighborsClassifier" class: Mahalanobis, Manhattan, Euclidean, and Cosine similarity. After preliminary testing, we selected Euclidean and Cosine distances for further analysis due to their high accuracy in our text classification tasks. Our evaluations included three detailed experiments: the first utilized the predefined KNN algorithm to calculate distances using the Euclidean metric, while the second applied Cosine similarity to better capture the complexities of our text data. In the third experiment, we implemented our CQKNN using the IBM quantum experience systems and its open-source Quantum Information Science Toolkit (Qiskit) framework to construct the quantum circuit depicted in Figure 5. As not all simulators and systems support the mid-circuit measurements feature, we used the "ibmq_cairo" back-end, which has 27 physical qubits, to ensure our setup was fully operational. The experimental setups include:

Table 1. The number of features used in our experiments, the number of qubits employed to encode *N*-dimensional feature vectors, and the total number of qubits employed in our CQKNN.

N features	$\log_2(N)$ qubits	$\log_2(N) \times 2 + 1$ qubits
8	3	7
16	4	9
32	5	11

1. Using a variety of features of our text dataset, as represented in **Table 1**.
   The table above delineates the scaling of quantum resource requirements with the increase in the dimensionality of feature vectors for our CQKNN implementation. It specifically details the number of qubits necessary to encode different sizes of feature vectors and the total qubits involved in the computation. This relationship is pivotal for optimizing quantum circuit efficiency and feasibility within the constraints of current quantum technology.
2. Perform experiments using different numbers of neighbors, specifically $k = \{1, 2, \ldots, 10\}$.
3. Compared to other QKNN, our CQKNN uses only one circuit and few qubits, as shown in **Table 2**.
   Our CQKNN method showcases a significant reduction in circuit complexity by employing only a single circuit for all computations, an innovative approach that contrasts sharply with other QKNN models. These models typically require a new circuit for each test and training sample comparison, often resulting in $M \times M'$ circuit configurations, which can be resource-intensive. Our method's efficiency is further highlighted in its non-reuse variant, which, while using the same number of qubits as the standard implementation, still necessitates $M \times M'$ circuits, mirroring the complexity of other QKNN approaches. This design optimizes the use of quantum resources, making our method more practical and feasible for current quantum technology, particularly on NISQ devices.
4. Our quantum circuit is constructed and subsequently executed using a transpiled version. The transpiled circuit allows for the direct implementation of quantum gates on the quantum hardware. The quantum circuit construction and transpilation implementation are carried out using IBM's open-source developed software package, Qiskit.
   The transpiler receives a logical quantum circuit **Figure 9a** as input and implements a series of transformations to generate

Table 2. Comparison of the number of qubit and circuit in our method with those of other methods, for *N*-dimensional features dataset with *M* training samples and *M'* test samples.

Algorithm	Total qubits to calculate distance	Circuit
Our method	$log_2(N) \times 2 + 1$	1
Our Method without Qubit Reuse	$log_2(N) \times 2 + 1$	$M \times M'$
QKNN[4]	$2(N-1) + 1$	$M \times M'$
QKNN[2,5]	$2 \times (N+3) + 1$	$M \times M'$
QKNN[6]	$log(M) + log_2(N) \times 2 + 1$	$M \times M'$

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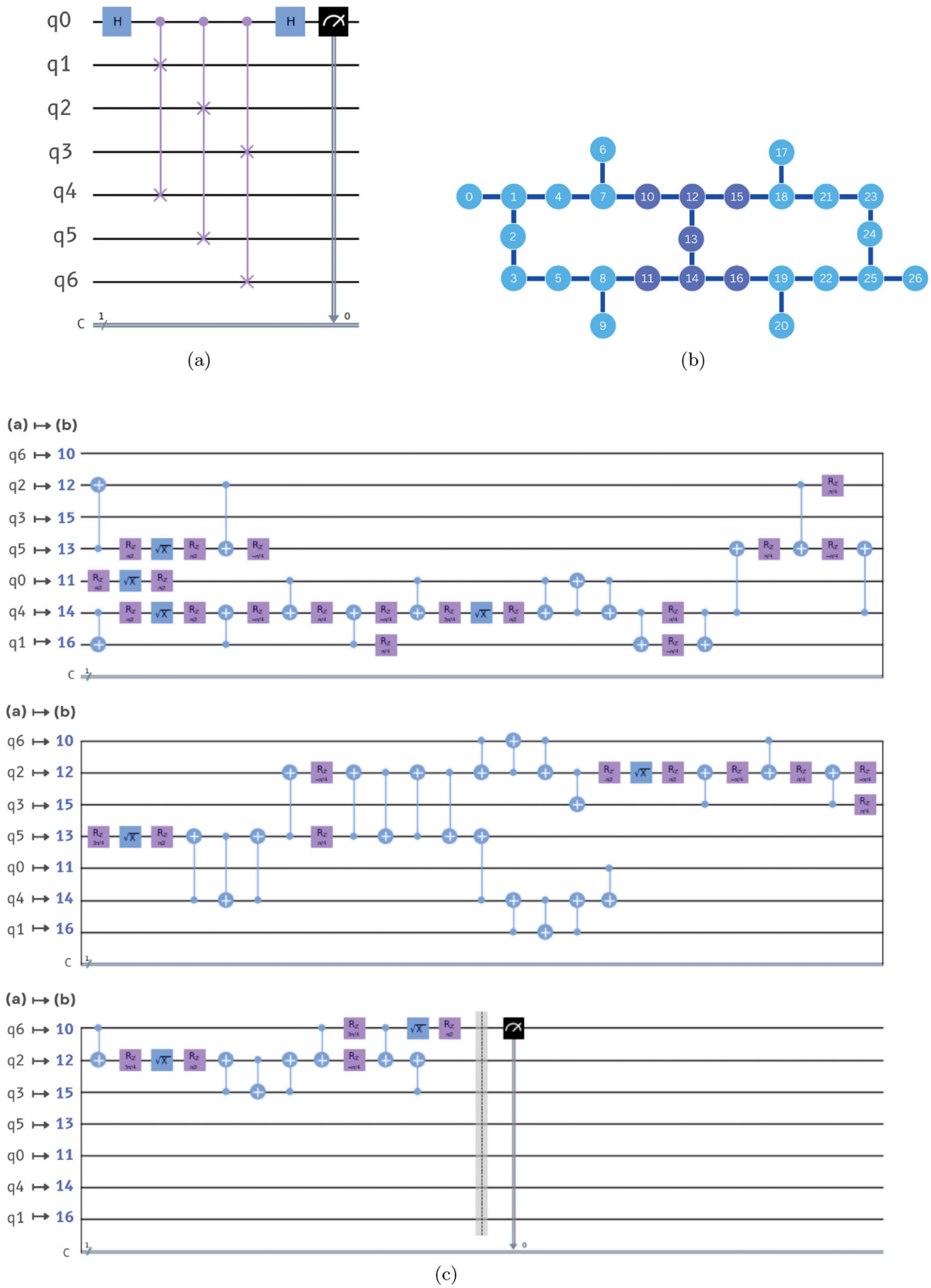


Figure 9. Example of the quantum transpilation stages of swap test in "ibmq_cairo" device. a) Input circuit before transpilation. b) The topology of the physical qubits. c) The physical circuit after transpilation.

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Table 3. Comparison of the initial and optimized depths of our CQKNN algorithm and its variant without qubit reuse in the SMS Spam dataset, applied to 50 samples with *N*-dimensional features. "Initial depth" indicates pre-transpilation depth, with subsequent columns showing depths across four optimization levels.

Method	N	Initial depth	Opt. Level 0	Opt. Level 1	Opt. Level 2	Opt. Level 3
Our method	8	3154	10 345	8763	5243	6705
	16	5150	14 465	11 230	7087	9167
	32	8045	17 578	15 067	9209	12 470
Our method without qubit reuse	8	12 215	21 579	16 782	13 347	14 233
	16	15 847	27 654	20 217	16 789	17 612
	32	20 308	33 129	27 153	21 347	23 176

a running circuit that conforms to the limitations of a particular device.

That device is characterized by a defined set of physical basis gates, including particular single qubit and also two-qubit gates. These physical gates are used to replace all logical gates present in the original circuit. In the current simulation, the number of physical gates is greater than that in the logical circuit. This is demonstrated in Figure 9c. The transpiler must consider the qubit connectivity of the quantum device to map logical qubits to physical qubits according to the topology of the physical qubits shown in Figure 9b, resulting in the creation of the physical circuit.

The physical qubits in the "ibmq_cairo" device, as depicted in Figure 9b, are not all interconnected. Consequently, the transpiler will attempt to assign the logical qubits to physical qubits while following their connectivity. In the initial circuit (Figure 9a), a controlled swap gate was applied to qubits q1 and q4. Consequently, the transpiler will attempt to assign these two logical qubits to two physically connected qubits (q1 to 16 and q4 to 14), and so on.

5. The transpiler employed four optimization levels (0 to 3) to reduce the circuit depth due to the increased number of physical gates resulting from transpilation. The depth of the circuit in Figure 9a is 6. However, when the circuit is transpiled with optimization_level = 2, the depth increases to 63. Without any optimization, the transpiled circuit has a depth of 87. **Table 3** outlines the quantum circuit depths for our CQKNN algorithm and its variant without qubit reuse, which is comparable to other QKNN models in the literature, across various optimization levels. This comparative analysis illustrates a substantial advantage in circuit complexity reduction achieved by our CQKNN method. Initially, the depths are considerably high due to the computational complexity inherent in quantum distance calculations. However, through strategic optimization, these depths are significantly decreased, particularly noticeable at optimization level 2 where we observe the most substantial reduction.

   This efficiency in reducing quantum circuit depth is critical as it directly correlates with fewer gate errors and less quantum decoherence, crucial for maintaining the integrity of quantum computations. Our method not only reduces the number of required gates but also effectively minimizes the quantum resource demands, enhancing both the feasibility and reliability of quantum computations on NISQ devices.

   On the other hand, implementing our CQKNN method without the qubit reuse feature is challenging due to significantly increased complexities from the higher number of circuits and gates. This requirement escalates quantum resource consumption and intensifies the challenges in maintaining quantum coherence over extended periods. This comparison underscores the practical benefits of our optimized CQKNN setup, which conserves quantum resources and offers a more feasible and reliable solution for executing complex machine-learning tasks on contemporary quantum systems. This table not only highlights the efficiency of our unique quantum circuit design but also emphasizes its practical implications in advancing quantum computing applications.

6. All experiments were conducted with the highest possible error mitigation provided by IBM quantum. Error mitigation techniques are employed to reduce circuit errors by modeling the hardware noise during execution. We employed three error mitigation methods provided by Qiskit runtime, as specified through the resilience_level option. These techniques encompass the following methods:
   - Twirled Readout Error eXtinction (T-Rex) with a resilience_level = 1: this approach mitigates measurement noise by incorporating additional measurement as well as adjustment circuits to estimate error-mitigated averages.[26]
   - Zero Noise Extrapolation (ZNE) with a resilience_level = 2: this method involves measuring a quantum observable at various noise levels achieved by artificially increasing hardware noise and then extrapolating the results to the limit of zero noise.[27]
   - Probabilistic Error Cancelation (PEC) with a resilience_level = 3: this technique commences with the execution of specific quantum circuits to identify imperfection models Subsequently random circuits receive samples from a distribution ensuring accurate computation results However due to the complexity of employing this advanced error mitigation strategy within the context of a single extensive circuit we did not conduct experiments as it would have necessitated additional resources.[27]
7. We conducted each experiment with the maximum allowed number of shots, which is 10^5, on the "ibmq_cairo" device. This was done to enhance the precision of the expectated results.

4.4. Results

In this subsection, we will compare the performance of three different models constructed using the SMS Spam Collection dataset, Twitter US Airline Sentiment dataset, and IMDB Movie Reviews dataset. Each dataset provides a unique context for evaluating the effectiveness of our models in real-world text classification scenarios.

Figure 10 demonstrates the enhanced performance of our CQKNN method with T-Rex and ZNE quantum error mitigation techniques compared to unmitigated scenarios across the SMS Spam Collection, Twitter US Airline Sentiment, and IMDB Movie Reviews datasets. Both techniques significantly outper-

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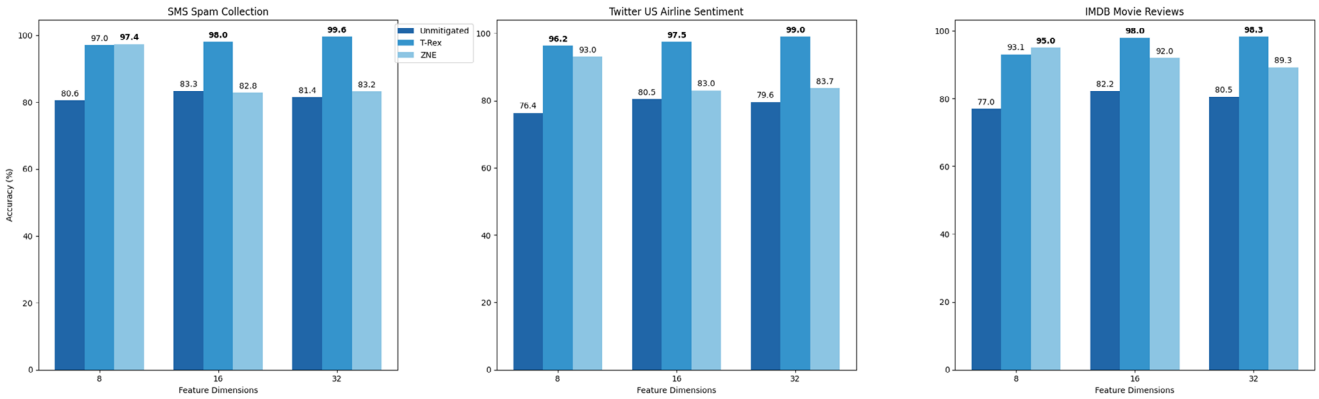


Figure 10. CQKNN prediction accuracy for all experiments, both with the use of error mitigation techniques and without any mitigation (unmitigated).

form the unmitigated approach, with T-Rex achieving up to 99.6% accuracy and ZNE excelling with 95.0% accuracy at specific feature dimensions.

As detailed in Figure 10, the integration of error mitigation techniques like ZNE and T-Rex not only improves the precision of our CQKNN model but also handles the challenges of increasing qubits and features. ZNE requires multiple executions under different noise conditions, enhancing accuracy, while T-Rex prepares a resilient noisy state that boosts operational efficiency. This analysis confirms the superiority of our quantum-enhanced methods over classical KNN, showcasing their effectiveness in complex text classification tasks and their potential to advance quantum machine learning applications on current NISQ devices.

Table 4 illustrates the superior accuracy of our CQKNN compared to traditional KNN using Euclidean distance and cosine similarity metrics across three datasets: SMS Spam Collection, Twitter US Airline Sentiment, and IMDB Movie Reviews. For instance, in the SMS Spam Collection, CQKNN achieves up to 99.6% accuracy with 32 features, significantly surpassing the traditional methods where cosine similarity and Euclidean distance peak at 92.3% and 80.5%, respectively. Similarly, CQKNN achieves 99.2% accuracy at 32 features for Twitter US Airline Sentiment and 99.4% for IMDB Movie Reviews, demonstrating its effectiveness in analyzing sentiment and nuanced textual data.

Table 4. Comparative accuracy of CQKNN and traditional KNN algorithms across multiple datasets and feature dimensions.

Dataset	Feature Dim.	CQKNN [%]	KNN Cos. [%]	KNN Euc. [%]
SMS Spam	8	97.4	88	70
	16	98	92	78
	32	99.6	92.3	80.5
Twitter US	8	96.2	90	72
	16	97.5	93	74.4
	32	99	92.3	82
IMDB Rev.	8	95	89.8	73
	16	98	91.6	73.8
	32	98.3	92	80.2

These results highlight the robust capabilities of CQKNN in delivering higher precision in text classification tasks.

The accompanying **Figure 11** plots the accuracy rates from the table, visually depicting the comparison between our CQKNN and traditional KNN algorithms across varying feature dimensions for each dataset. This visualization highlights the stark contrasts in performance, with CQKNN consistently outperforming the others, especially at higher feature dimensions, emphasizing its suitability for complex text classification scenarios.

Figure 12 illustrates the comparative performance analysis of accuracy rates for different values of k using the SMS Spam Collection dataset. This analysis contrasts the CQKNN algorithm with traditional KNN methods employing Euclidean distance and cosine similarity metrics. The focus on this specific dataset is due to the exemplary performance exhibited by our CQKNN algorithm in this context. The graph demonstrates that CQKNN consistently achieves superior accuracy, maintaining near-optimal levels across various k values, markedly outpacing the performance of traditional methods. In contrast, KNN using Euclidean distance displays considerable variability, particularly at lower k values, while KNN using cosine similarity, although relatively stable, fails to match the high accuracy levels achieved by CQKNN. This visual representation highlights the robustness of our quantum-enhanced approach in effectively managing the intricacies of spam detection and underscores its potential for significant improvements in accuracy within text classification tasks.

5. Conclusion and Perspectives

To conclude, this paper introduces a novel hybrid CQKNN algorithm, which employs a unique quantum circuit design to achieve superior functionality compared to existing QKNN algorithms. The incorporation of the qubit reuse technique enables our CQKNN to efficiently handle a substantial volume of data points without the need for an extensive number of qubits by utilizing just a single circuit and decreasing the number of gates. This presents a significant benefit to our methodology. The optimized structure of our circuit design not only improves computational efficiency but also positions our algorithm as a promising solution for practical applications in quantum-enhanced machine learning.

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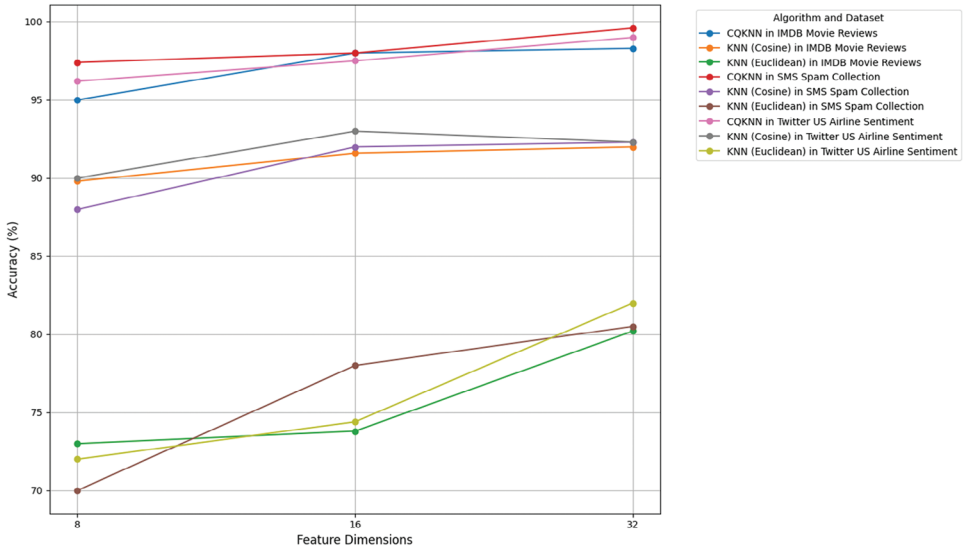


Figure 11. The classification accuracy of KNN using Euclidean distance, KNN using Cosine similarity, and our proposed method CQKNN based on the feature dimension across three datasets.

In our research study, we applied our hybrid QKNN algorithm to the task of classifying text within three key datasets: the SMS Spam Collection, Twitter US Airline Sentiment, and IMDB Movie Reviews datasets. The CQKNN method exhibited superior performance across these diverse contexts, particularly shining in the SMS Spam Collection dataset where it significantly outperformed traditional KNN methods. This performance boost is attributed to the practical implementation of fidelity as a measure of similarity, transitioning from theoretical applications to actual quantum computations, which enhanced accuracy and demonstrated the practical potential of quantum techniques in real-world applications.

Moreover, the application of two quantum error mitigation techniques, T-Rex and ZNE, effectively reduced computational noise, further boosting the reliability and accuracy of our model. These results underscore the robustness of our CQKNN approach in harnessing quantum computing's potential for complex data classification tasks, marking a significant stride toward operational quantum machine learning systems.

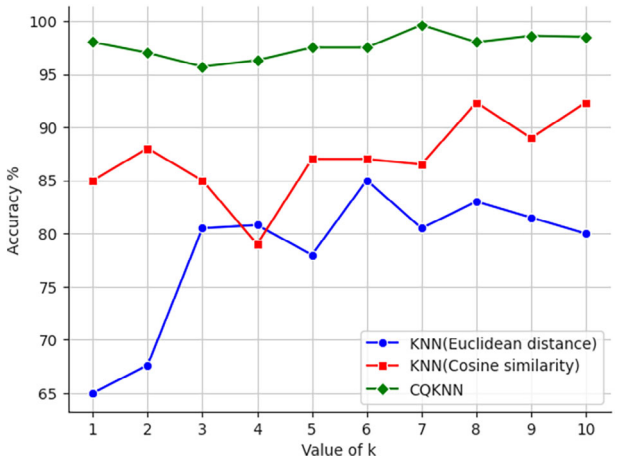


Figure 12. Classification accuracy comparison of KNN with Euclidean distance, Cosine similarity, and our CQKNN in the SMS Spam Collection dataset, evaluated across feature dimensions 8, 16, and 32 for varying values of k.

In our continuous and future study goals, our commitment is in expanding the current limits of quantum machine learning. Our main goal is to advance our methodology toward a purely quantum K-Nearest Neighbors algorithm. This algorithm will incorporate quantum techniques throughout its entirety, including the utilization of a quantum algorithm for identifying the minimum value and the Grover algorithm to search k values. Additionally, our aim is to ensure that this algorithm is capable of efficiently processing feature vectors of considerable size. The previously mentioned sophisticated method exhibits promise for practical implementation, especially in the field of text recommendation tasks, where the quantum advantage may be optimized. Nevertheless, the achievement of this objective relies heavily on the broader accessibility of physical quantum computing resources that have been provided with a significant quantity of qubits. We remain optimistic that as quantum technology continues to advance, greater accessibility to such resources will become a reality for researchers. This, in turn, will facilitate the development of more resilient and efficient applications of quantum machine learning.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Keywords

hybrid classical quantum k-nearest neighbors, quantum computing, quantum machine learning, qubit reuse, text classification

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[1] M. A. Nielsen, I. L. Chuang, *Quantum computation and quantum information*, Cambridge University Press, Cambridge **2010**.
[2] N. Wiebe, A. Kapoor, K. Svore, *(preprint) arXiv:1401.2142*, **2014**.
[3] Y. Ruan, X. Xue, H. Liu, J. Tan, X. Li, *Int. J. Theor. Phys.* **2017**, *56*, 3496.
[4] V. T. Hai, P. H. Chuong, *Informatica* **2022**, *46*, 5.
[5] L. Z. Gao, C. Y. Lu, G. D. Guo, X. Zhang, S. Lin, *Front. Phys.* **2022**, *10*, 1047466.
[6] C. Feng, B. Zhao, X. Zhou, X. Ding, Z. Shan, *Entropy* **2023**, *25*, 127.
[7] F. Edward, H. Neven, *(preprint) arXiv:1802.06002*, **2018**.
[8] N. Killoran, T. R. Bromley, J. M. Arrazola, M. Schuld, N. Quesada, S. Lloyd, *Phys. Rev. Res.* **2019**, *1*, 033063.
[9] I. Cong, S. Choi, M. D. Lukin, *Nat. Phys.* **2019**, *15*, 1273.
[10] P. Rebentrost, M. Mohseni, S. Lloyd, *Phys. Rev. Lett.* **2014**, *113*, 130503.
[11] R. Zhang, J. Wang, N. Jiang, H. Li, Z. Wang, *Neural Networks* **2022**, *151*, 376.
[12] G. Wang, *Phys. Rev. A* **2017**, *96*, 012335.
[13] C. H Yu, F. Gao, Q. Y. Wen, *IEEE Trans. Knowl. Data Eng.* **2019**, *33*, 858.
[14] S. Lloyd, M. Mohseni, P. Rebentrost, *Nat. Phys.* **2014**, *10*, 631.
[15] L. E. Peterson, *Scholarpedia* **2009**, *4*, 1883.
[16] G. Guo, H. Wang, D. Bell, Y. Bi, K. Greer, *Soft Comput.* **2006**, *10*, 423.
[17] L. H. Gong, W. Ding, Z. Li, Y. Z. Wang, N. R. Zhou, *Adv. Quantum Technol.* **2024**, 2300221.
[18] M. Hdaib, S. Rajasegarar, L. Pan, *Quant. Mach. Intell.* **2024**, *6*, 26.
[19] A. Zeguendry, Z. Jarir, M. Quafafou, *Entropy* **2023**, *25*, 287.
[20] Wikipedia. Quantum Machine Learning, https://en.wikipedia.org/w/index.php?title=Quantum_machine_learning&oldid=1055622615 (accessed: January 2024).
[21] H. Buhrman, R. Cleve, J. Watrous, R. De Wolf, *Phys. Rev. Lett.* **2001**, *87*, 167902.
[22] M. DeCross, E. Chertkov, M. Kohagen, M. Foss-Feig, *Phys. Rev. X* **2023**, *13*, 041057.
[23] T. Almeida, J. M. Hidalgo, T. Silva, *Int. J. Inf. Secur. Sci.* **2013**, *2*, 1.
[24] Twitter US Airline Sentiment, https://www.kaggle.com/datasets/crowdflower/twitter-airline-sentiment (accessed: January 2024).
[25] A. L. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, C. Potts, *Proc. Assoc. Comput. Linguist.* **2011**, 142.
[26] E. Van Den Berg, Z. K. Minev, K. Temme, *Phys. Rev. A* **2022**, *105*, 032620.
[27] K. Temme, S. Bravyi, J. M. Gambetta, *Phys. Rev. Lett.* **2017**, *119*, 180509.

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